Rubber_band_analysis

Analysis of Field Data and AI-AI in Rubber-band dynamics

PENPAL Project

2026-01-09

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0.1 Summary

- Analysis of field data (Human-AI) reveals a robust "rubber-band" effect: difference between valence gaps in one turn strongly predict opposite-direction changes in the next turn.
- This effect is seen when measuring valence gap relative to both the AI's current turn and its previous turn.
- The effect is also present when using cosine distance between user and AI embeddings as the alignment metric.
- This indicates that emotional divergence is temporary and snaps back toward alignment.
- There is an intrinsic stabilising tendency in the co-creation process, ensuring narrative cohesion.

1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 870 17
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"                 "user"
##  [3] "ai"                        "conversation_id"
##  [5] "respondent_id"             "interaction_count"
##  [7] "llm_type"                  "edit_distance"
##  [9] "turn"                      "starter"
## [11] "user_embedding"            "ai_embedding"
## [13] "pct_turn"                  "user_sentiment_score"
## [15] "ai_sentiment_score"        "user_sentiment_projection"
## [17] "ai_sentiment_projection"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1              87     783                    9           -0.0831         0.00386
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===
```

```
## Field data - Turn 1 with non-NA lag (should be 0): 0
```

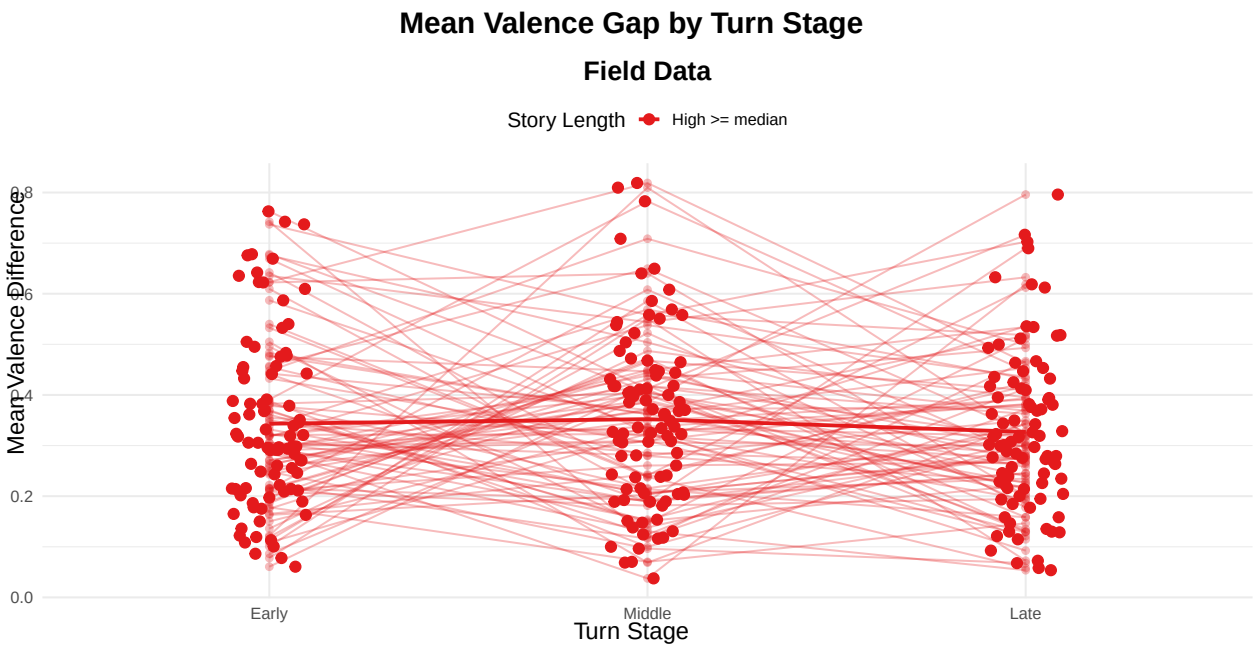
2 2. Temporal Dynamics Analysis

2.1 2.1 Stage-Based Analysis (Early/Middle/Late)

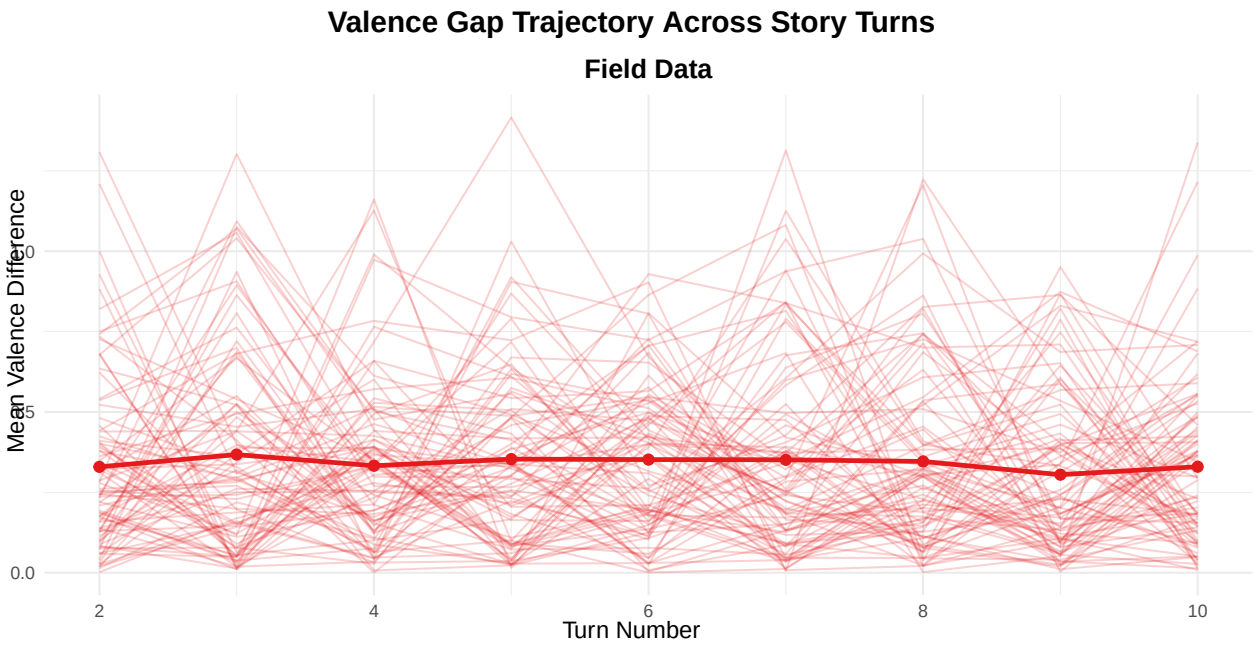
```
## === Mean Valence Gap by Stage ===
```

```
## # A tibble: 3 x 4
##   data_source stage_label mean_gap sd_gap
##   <chr>       <fct>          <dbl>  <dbl>
## 1 Field       Early          0.343     NA
## 2 Field       Middle         0.352     NA
## 3 Field       Late           0.327     NA
```

2.2 2.2 Stage Valence Gap Plot



2.3 2.3 Turn-by-Turn Valence Gap Trajectory



2.4 2.4 Rubber-Band Dynamics

```
## === Rubber-Band Model: Field Data ===
```

```
##                Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -0.07987795 0.05283233 -1.511914 1.342648e-01
## delta_12    -0.49037167 0.08246523 -5.946405 5.895343e-08
```

```
##
## === Rubber-Band Model: Simulated Data ===
```

2.5 2.5 Turn-by-Turn Rubber-Band Dynamics

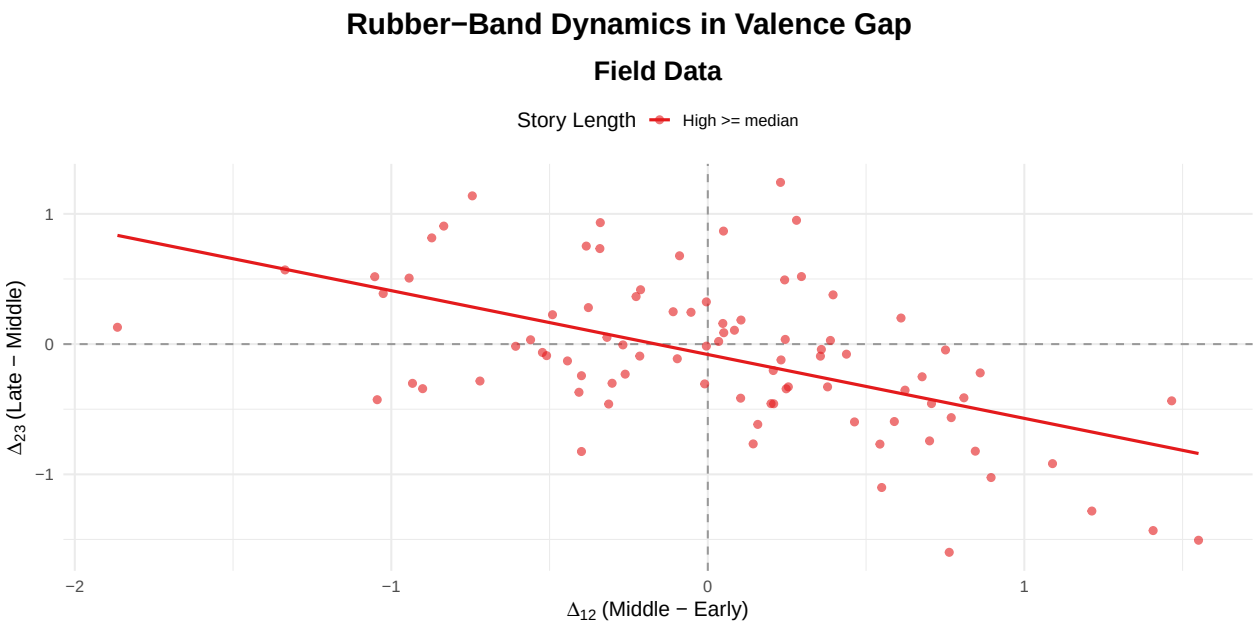
```
## === Direction Turn-Level Rubber-Band Model: Field Data ===
```

```
##                         Estimate Std. Error   df      t value     Pr(>|t|)
## (Intercept)         -0.021663029 0.03780826 1127  -0.57297086 5.667787e-01
## gap_change          -0.462101380 0.03767626 1127 -12.26505507 1.497642e-32
## typesame             0.001106487 0.05152375 1127   0.02147529 9.828703e-01
## gap_change:typesame -0.003300879 0.05242142 1127  -0.06296814 9.498030e-01
```

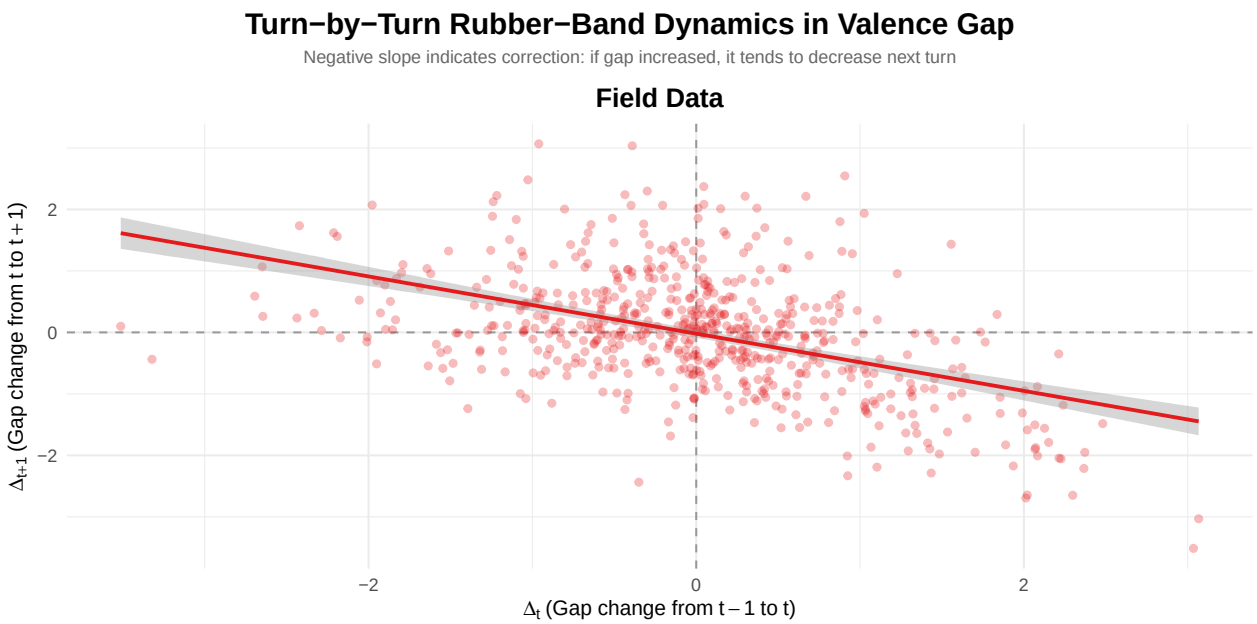
```
##
##
## === Turn-Level Rubber-Band Correlation ===
```

```
## Field correlation (gap_change vs gap_change_next): -0.4673788
```

2.6 2.6 Rubber-Band Dynamics Plot



2.7 2.7 Turn-Level Rubber-Band Dynamics Plot



3 3. Permutation tests

3.1 3.1 Stage-level Permutation Test

```
## === Stage-Level Permutation Test Results ===
## Observed slope: -0.4904
## Null mean: -0.5004
## Null SD: 0.0964
## Effect size (z): 0.1
## P-value: 0.526
```

3.2 3.2 Brutal Stage-Level Permutation

```
## === Stage-Level Permutation Test Results - Shuffle Turn-Level Valence Gap first ===
## Observed slope: -0.4904
## Null mean: -0.5054
## Null SD: 0.0938
## Effect size (z): 0.16
## P-value: 0.566
```

3.3 3.2 Global Shuffle Stage-Level Test

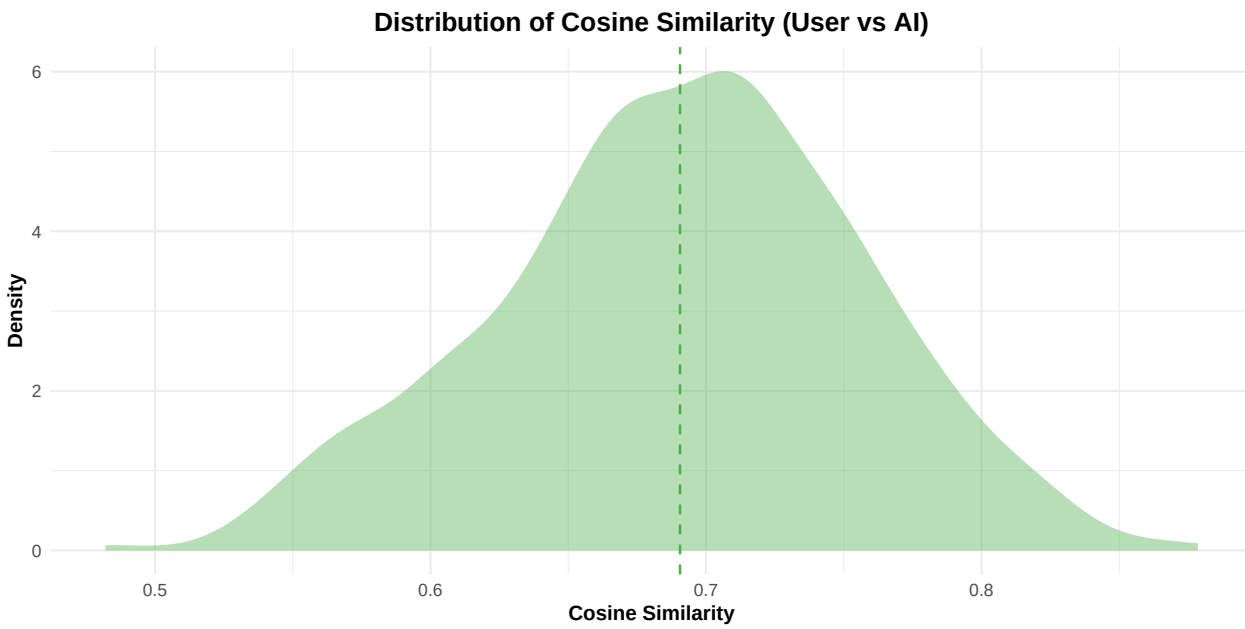
```
##                Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)  0.09702642 0.04589683  2.114011 0.04389397
## d12         -0.39310856 0.20716272 -1.897583 0.06849270
```

3.4 3.3 Turn-Level Permutation Test

```
## === Turn-Level Permutation Test Results ===

## Observed slope: -0.4654

## Null mean: 0.001

## Null SD: 0.0519

## Effect size (z): -8.99

## P-value: <2e-16

## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.500      -0.509   174
## 2 Within-Stage      -0.442      -0.457   348
```

4 4. Cosine similarity alignment

4.1 4.1 Cosine Similarity Distribution

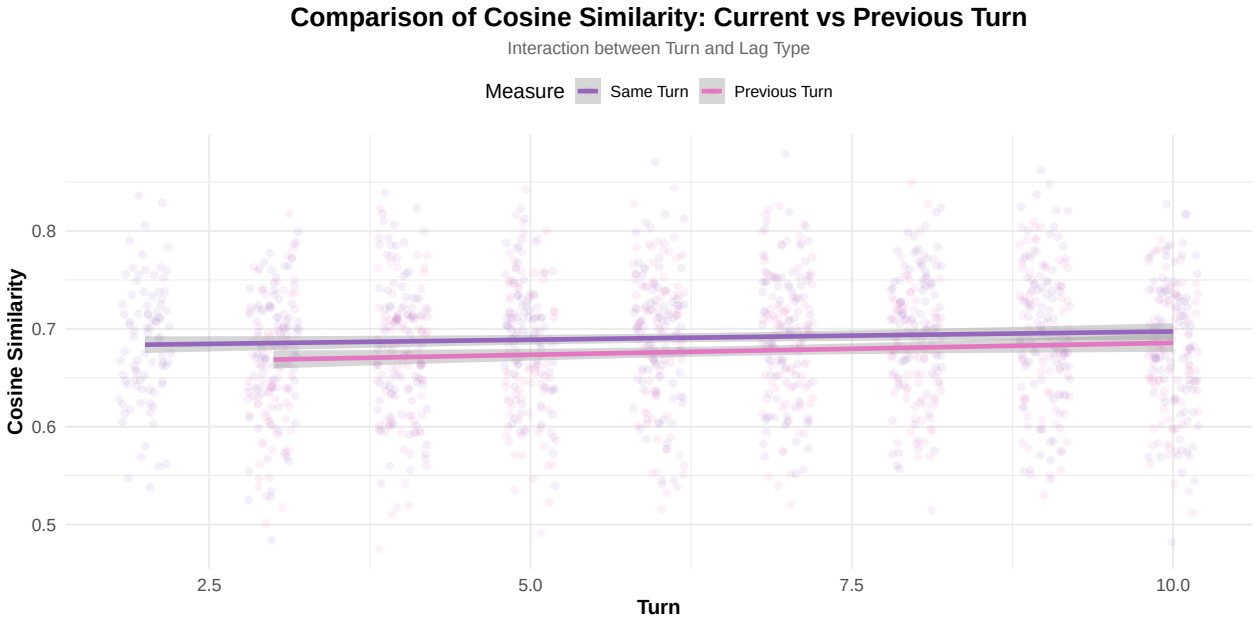


4.2 4.2 Cosine Similarity Mixed Model (Interaction)

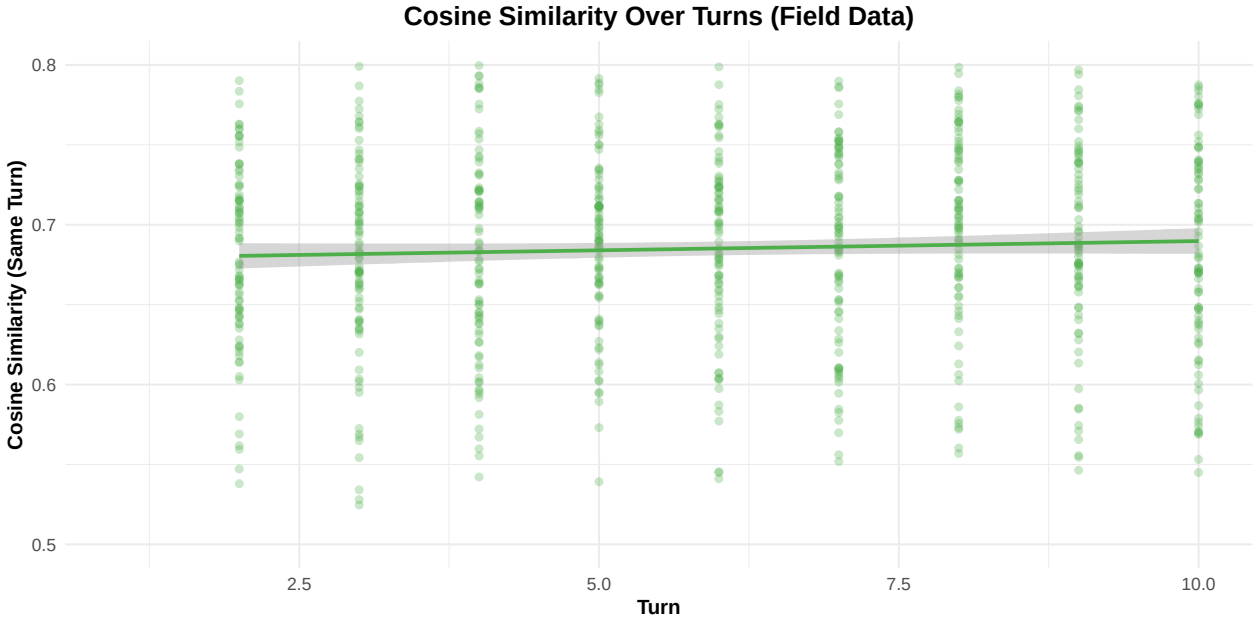
```
## === Cosine Similarity Interaction Model (Same vs Previous Turn) ===

##                                Estimate   Std. Error        df     t value
## (Intercept)                 0.6804414210 0.0062392872  796.4603 109.0575575
## turn                        0.0016875825 0.0008494414 1389.0000   1.9866969
## lag_typePrevious Turn      -0.0191162458 0.0089302030 1389.0000  -2.1406284
## turn:lag_typePrevious Turn  0.0007526789 0.0013237588 1389.0000   0.5685922
##                              Pr(>|t|)
## (Intercept)                0.00000000
## turn                       0.04715221
## lag_typePrevious Turn      0.03247771
```

```
## turn:lag_typePrevious Turn 0.56972486
```

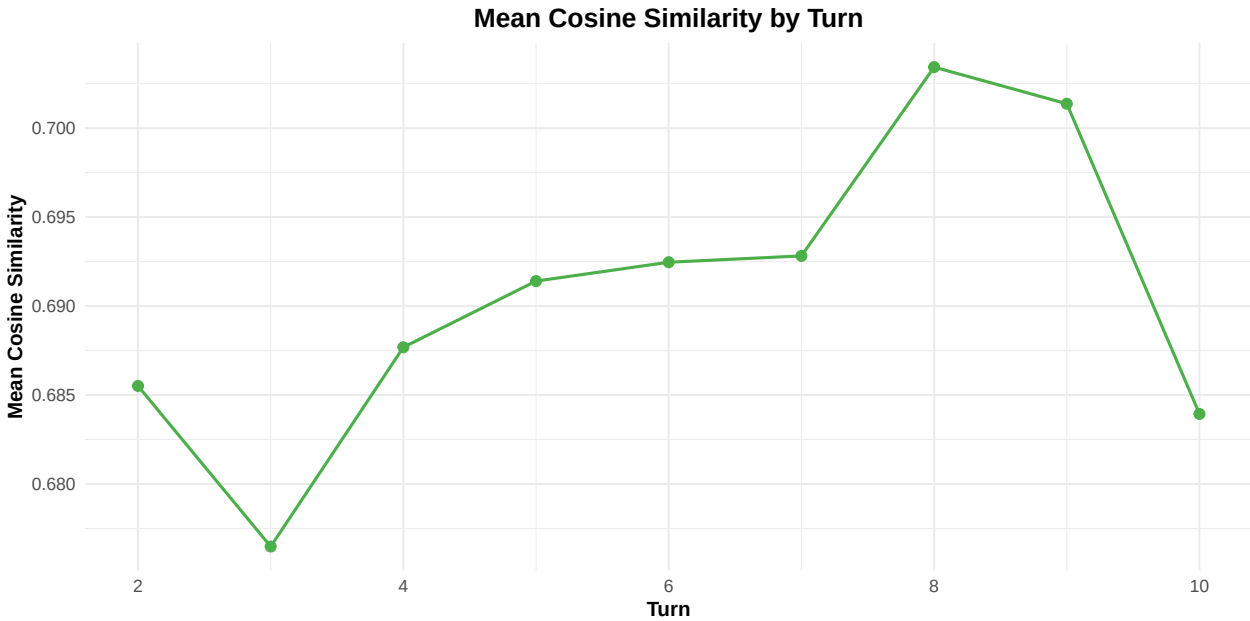
4.3 4.2b Comparison of Current vs Previous Turn Cosine Similarity

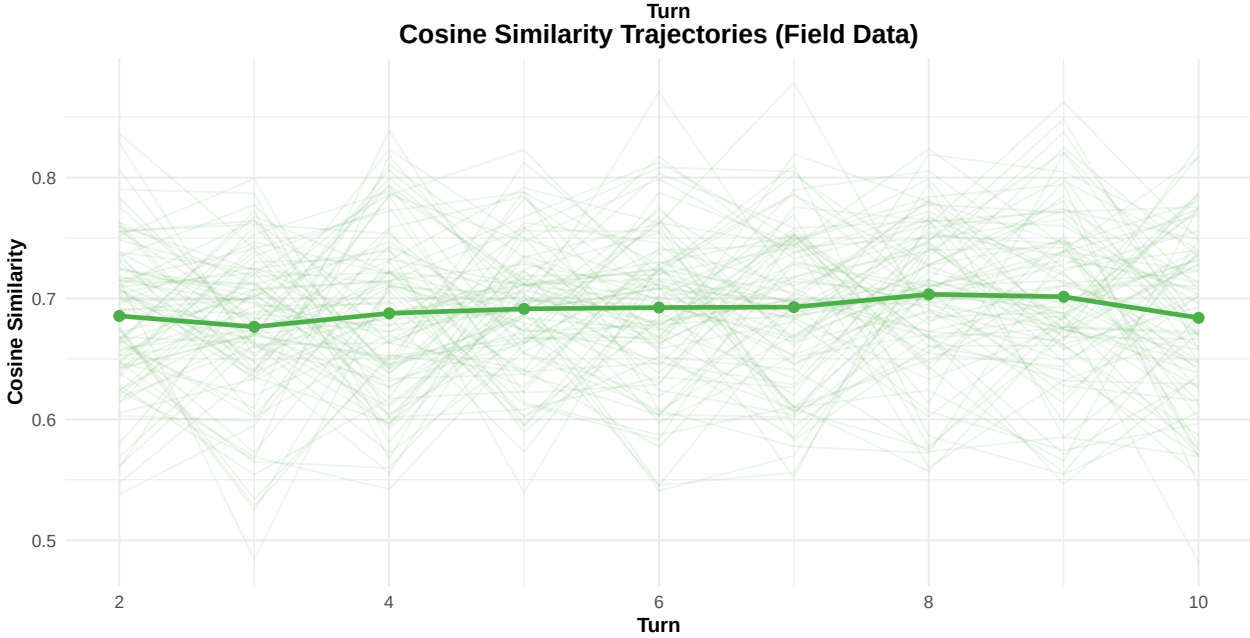


4.4 4.3 Cosine Similarity Linear Plot



4.5 4.4 Cosine Similarity Over Time





4.6 4.5 Cosine rubber band

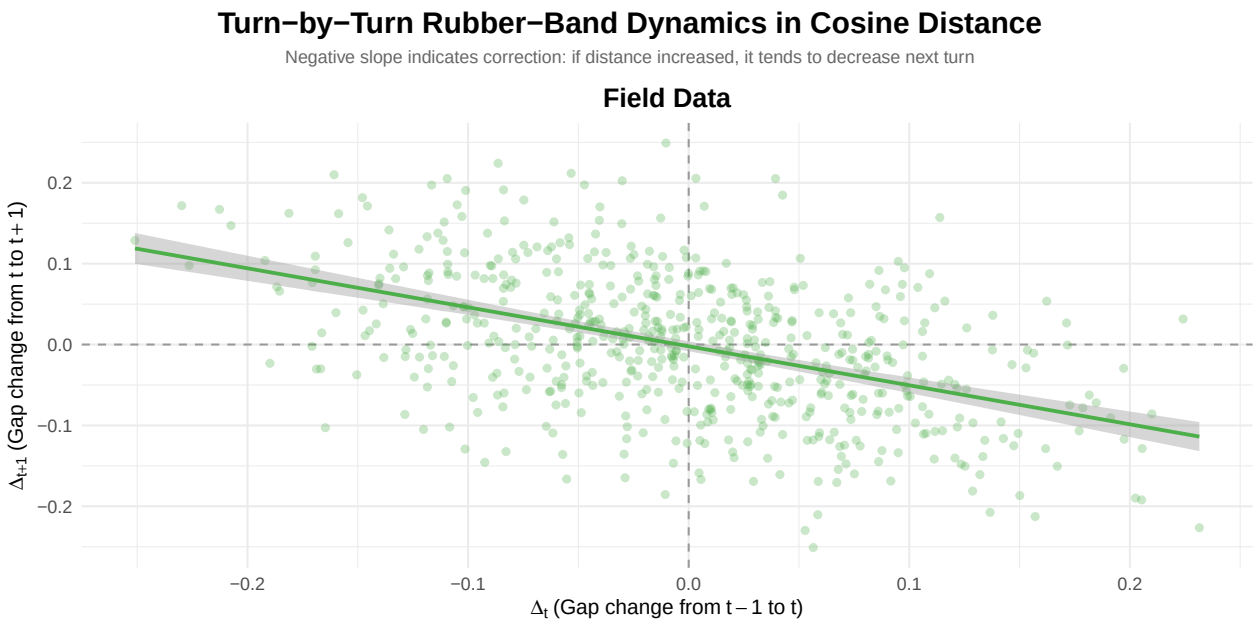
```
## === Turn-Level Rubber-Band Cosine Distance: Field Data ===

##                 Estimate Std. Error     t value     Pr(>|t|)
## (Intercept) -0.002157011 0.00311352  -0.6927884 4.887073e-01
## dist_change -0.482254060 0.03680596 -13.1026091 1.015276e-34
```

4.7 4.6 Turn level Cosine Permutation test

```
## === Cosine Distance Turn-Level Permutation Test Results ===

## Observed slope: -0.4823

## Null mean: 0.003
```

```
## Null SD: 0.0515
## Effect size (z): -9.43
## P-value: <2e-16
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.470      -0.485   174
## 2 Within-Stage      -0.487      -0.488   348
```

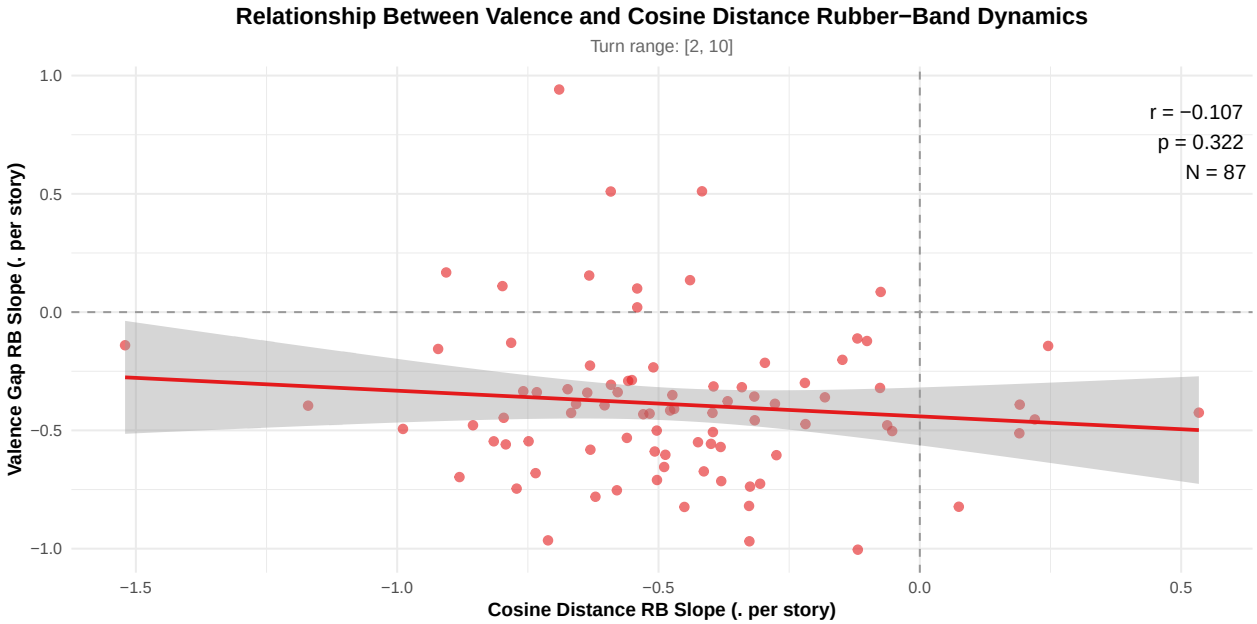
4.8 4.7 Turn Level Cosine RB Linear plot



4.9 4.8 Correlation Between Cosine Distance and Valence Rubber-Band

```
## === Correlation Between Valence and Cosine Distance Rubber-Band ===
## Turn range: [2, 10]
## N observations (turn-level): 609
## Turn-level correlation of gap changes (Δt): 0.085
##   Interpretation: Do valence and cosine gaps change in the same direction
##                   at the same turn across all stories?
## Turn-level correlation of next changes (Δt+1): 0.103
##   Interpretation: Do subsequent valence and cosine corrections correlate?
## N stories with >= 3 turns: 87
## N stories with >= 3 turns: 87
## STORY-LEVEL correlation of RB slopes: -0.107
##   Interpretation: Across stories, does valence RB strength predict cosine RB strength?
##                   Positive r means stories with strong valence correction
```

```
##                    also show strong cosine distance correction.
##   95% CI: [-0.311, 0.106]
##   p-value: 0.322
##   df: 85
## Shared variance (R²): 0.012
## Interpretation: 1.2 % of variation in valence RB slope
##                 is explained by cosine distance RB slope
```




```
##
## === Regression: Cosine RB Predicting Valence RB ===
##                   Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)     -0.4407742 0.06152664 -7.1639575 2.638646e-10
## cosine_rb_slope -0.1085049 0.10901657 -0.9953069 3.224120e-01
##
## === VALIDATION CHECK ===
## Correlation from cor(): -0.107
## Correlation from lm R²: 0.107
## These should match (they do if correlation is positive).
```

5 5. Supplementary: Interaction Effects

5.1 5.1 Starter Interaction Effects

5.1.1 5.1.1 Valence Gap by Starter

5.1.2 5.1.2 Cosine Distance by Starter

5.1.3 5.1.3 Alignment (Cosine Similarity) by Starter

5.2 5.2 LLM Type Interaction Effects

5.2.1 5.2.1 Valence Gap by LLM Type

```
## === Valence Rubber-Band: LLM Type Interaction ===
```

```
##                                 Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)                   0.03951076 0.07570498  0.5219044 6.019292e-01
## gap_change                   -0.45145613 0.08280982 -5.4517221 7.291431e-08
## factor(llm_type)2            -0.12504050 0.10435100 -1.1982683 2.312849e-01
## factor(llm_type)3            -0.02546250 0.09846390 -0.2585974 7.960345e-01
## factor(llm_type)4            -0.08668151 0.10105394 -0.8577747 3.913588e-01
## gap_change:factor(llm_type)2  0.04318321 0.11370882  0.3797701 7.042502e-01
## gap_change:factor(llm_type)3 -0.06092521 0.10632121 -0.5730297 5.668389e-01
## gap_change:factor(llm_type)4 -0.02363765 0.10476197 -0.2256320 8.215643e-01
```

5.2.2 5.2.2 Cosine Distance by LLM Type

```
## === Cosine Distance Rubber-Band: LLM Type Interaction ===
```

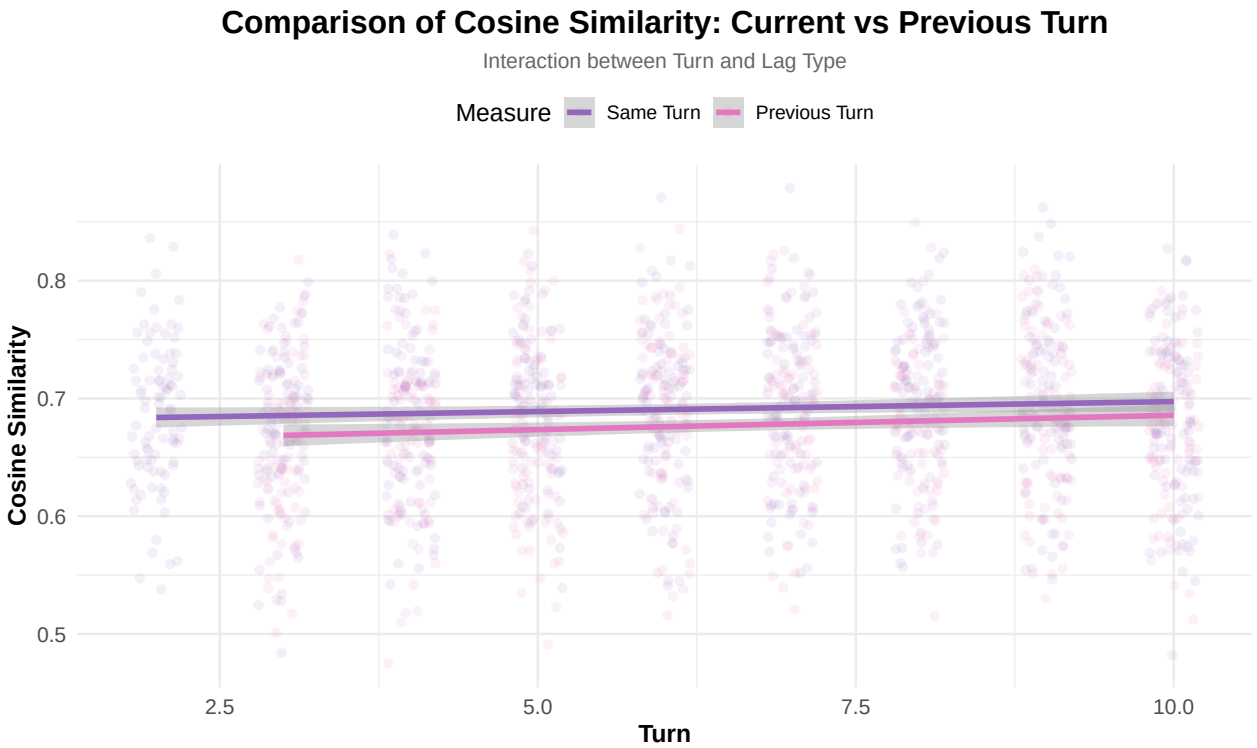
```
##                                    Estimate  Std. Error  df     t value
## (Intercept)                   -0.0014564580 0.006853238 601 -0.21252116
## dist_change                   -0.5352315824 0.074209467 601 -7.21244341
## factor(llm_type)2             -0.0006001643 0.009445964 601 -0.06353659
## factor(llm_type)3             -0.0035290171 0.008922452 601 -0.39552101
## factor(llm_type)4              0.0015968758 0.009149432 601  0.17453277
## dist_change:factor(llm_type)2  0.1868080551 0.107521805 601  1.73739694
## dist_change:factor(llm_type)3  0.0101079823 0.099602912 601  0.10148280
## dist_change:factor(llm_type)4  0.0334327835 0.108024028 601  0.30949395
##                                   Pr(>|t|)
## (Intercept)                   8.317725e-01
## dist_change                   1.660537e-12
## factor(llm_type)2             9.493603e-01
## factor(llm_type)3             6.925988e-01
## factor(llm_type)4             8.615056e-01
## dist_change:factor(llm_type)2 8.282941e-02
## dist_change:factor(llm_type)3 9.192011e-01
## dist_change:factor(llm_type)4 7.570531e-01
```

5.2.3 5.2.3 Alignment (Cosine Similarity) by LLM Type

```
## === Cosine Similarity (Alignment): LLM Type × Turn Interaction ===
```

```
##                             Estimate  Std. Error       df    t value
## (Intercept)             0.6722860138 0.013270809 618.7203 50.6590057
## turn                    0.0018940861 0.001899431 692.0000  0.9971860
## factor(llm_type)2       0.0146758600 0.018292549 618.7203  0.8022862
## factor(llm_type)3       0.0138340105 0.017263827 618.7203  0.8013293
## factor(llm_type)4       0.0024486500 0.017718438 618.7203  0.1381978
```

```
## turn:factor(llm_type)2 -0.0004542748 0.002618185 692.0000 -0.1735075
## turn:factor(llm_type)3  0.0008212035 0.002470946 692.0000  0.3323438
## turn:factor(llm_type)4 -0.0013144179 0.002536014 692.0000 -0.5183008
##                            Pr(>|t|)
## (Intercept)            2.541433e-222
## turn                    3.190228e-01
## factor(llm_type)2       4.226954e-01
## factor(llm_type)3       4.232487e-01
## factor(llm_type)4       8.901290e-01
## turn:factor(llm_type)2  8.623033e-01
## turn:factor(llm_type)3  7.397303e-01
## turn:factor(llm_type)4  6.044141e-01
```




```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: valence_gap ~ turn * lag_type + (1 | conversation_id)
##    Data: df_valence_long
##
## REML criterion at convergence: 331.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.8276 -0.7685 -0.1778  0.5825  4.6341
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.005066 0.07118
##  Residual                    0.068508 0.26174
## Number of obs: 1479, groups:  conversation_id, 87
##
## Fixed effects:
```

```
##                              Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                 3.569e-01  2.486e-02  1.205e+03  14.356   <2e-16
## turn                       -2.694e-03  3.623e-03  1.389e+03  -0.744    0.457
## lag_typePrevious Turn       2.968e-02  3.809e-02  1.389e+03   0.779    0.436
## turn:lag_typePrevious Turn -3.426e-04  5.646e-03  1.389e+03  -0.061    0.952
##
## (Intercept)                ***
## turn
## lag_typePrevious Turn
## turn:lag_typePrevious Turn
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) turn   lg_tPT
## turn        -0.874
## lg_typPrvsT -0.591  0.571
## trn:lg_tyPT  0.561 -0.642 -0.933
```